Structure-Based Druggability of Lung Adenocarcinoma Targets: FGFR2 and CYP1A1

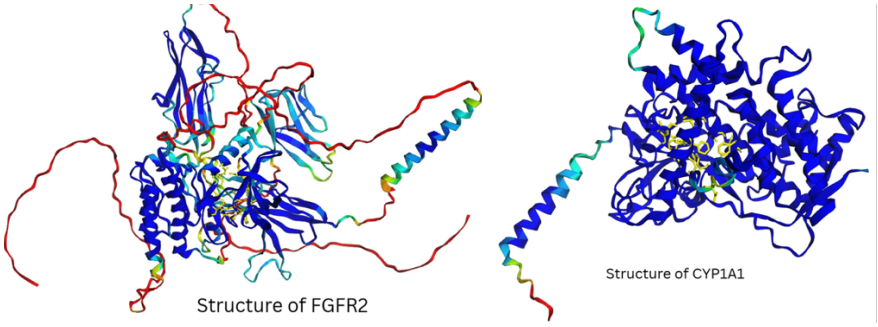
BIOT 6900 · Module 4 Assignment

Overview

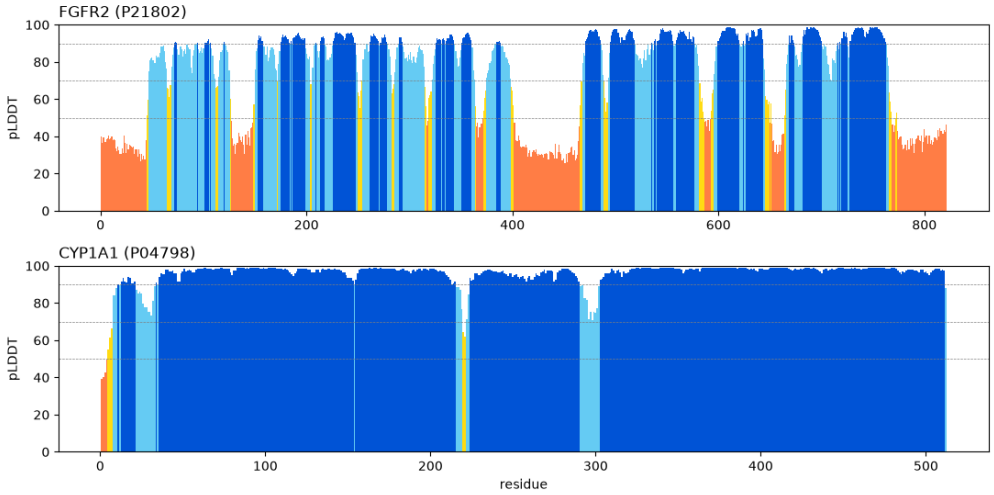
This assignment asks a question my earlier modules could not answer: do two of my LUAD targets, FGFR2 and CYP1A1, actually have a real physical pocket a drug could bind, not just whether they resemble known drug targets by sequence. I used AlphaFold predictions, checked how confident each region of the structure is, compared both against real lab-solved structures, and searched for genuine drug-shaped pockets. AI-assisted help wrote and debugged the Python; the biology reasoning and final calls below are mine. BRCA2, my original third choice coming out of Module 3, was swapped for CYP1A1 since its full protein is over 3,400 residues, well past this assignment's 2,700-residue limit.

Summary of key results:

Gene	Mean pLDDT	Domains	PDB used	RMSD (confident)	Best pocket score
FGFR2	73.9	5	3RI1 (kinase domain)	0.65 Å	0.882
CYP1A1	95.5	1	4I8V	0.33 Å	0.991



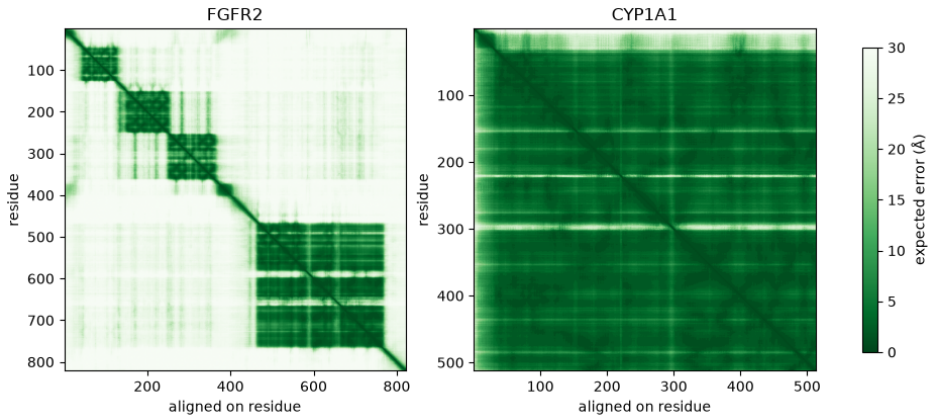
Which parts of each target can I trust



pLDDT per residue. Dark blue = very high, orange = very low.

CYP1A1 is solid dark blue throughout, no floppy regions anywhere, exactly what you would expect from one compact, single-domain enzyme. FGFR2 tells a different story: its orange blocks are real, flexible linker regions connecting its separate extracellular,

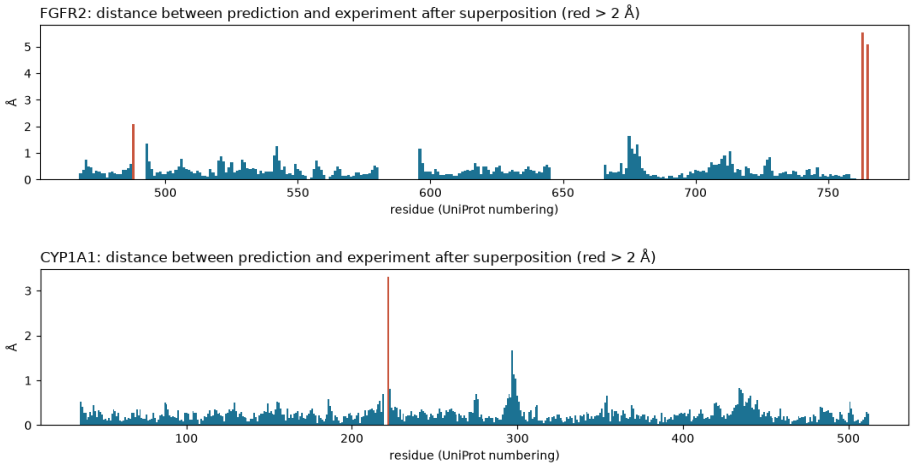
membrane-spanning, and kinase pieces, not a failed prediction, since a multi-domain receptor naturally needs some give between its parts.



PAE: dark = confident relative position, pale = uncertain.

FGFR2 splits into five confident domains, but the gap between the first two carries over 27 Å of expected error, so I only analyzed pockets within one domain at a time. CYP1A1 is one reliably-arranged domain.

How well do the predictions match real experimental structures



Distance between prediction and experiment after superposition. Red exceeds 2 Å.

The crystal structure of CYP1A1 was in perfect agreement with the real one, showing an RMSD of 0.33 Å among 475 confident residues. However, only one domain of FGFR2 could be found with a matching real crystal, the one of the kinase domain, which had an RMSD of 0.65 Å among 256 residues. It should be noted that only the kinase domain of the real FGFR2 has been crystalized.

Where they do differ, it is mostly because the crystal has something the prediction lacks. In FGFR2 the largest deviations were residues 763 and 765 (5.5 and 5.1 Å), at the very end of the crystallised kinase construct, where loose ends are expected, and residue 488 (2.1 Å), which lines the inhibitor's binding site in 3RI1. In CYP1A1 the largest deviation was residue 222 (3.3 Å), which is also one of the residues touching the bound inhibitor. So these differences reflect the crystal having a drug bound that pushes nearby loops into place, not errors in the prediction.

Does the structure actually support a drug pocket

Both proteins produced one strong, confidence-backed pocket after filtering, and in both cases that pocket turned out to be exactly where a real drug physically sits in the lab-solved structure. CYP1A1's pocket scored 0.991 out of 1 and covered 94% of its real

inhibitor's binding site and 93% of where its natural heme cofactor sits, two independent real molecules landing in the same one spot. FGFR2's pocket scored 0.882 and covered 100% of its real inhibitor's site, full overlap.

Tellingly, FGFR2's second-best pocket by raw geometry failed the confidence check outright, likely sitting in a floppy region, and a third was flagged suspect for unreliable relative positioning. Without those checks I would have had two or three candidates; with them, exactly one trustworthy answer per protein, matching the real drug site both times.

My call for both targets: structure supports a small-molecule pocket, backed by geometry, confidence, and direct ligand confirmation.

How does this compare with my earlier rankings

FGFR2 has outperformed at every stage, from the multi-omics ranking, to Module 3 tractability, and now a physical pocket that has been verified, a combination of three very different techniques that agree on the importance of the gene. CYP1A1 ranked eighth in Module 2, but now has the most structurally clean result of the two targets. I would prioritize both, with FGFR2 being the main target, while CYP1A1 has proven itself to pass every stage.

The reason why this consistency is so significant is because each stage evaluated a fundamentally different piece of information. In Module 2, the question was whether a gene has been shown to be dysfunctional in real patient data. In Module 3, the question was whether the sequence of a protein resembles something that can be drugged already, which is a pattern matching argument without any physical proof behind it. In this module, the question was whether the protein has a physical location for a drug to bind to. Having consistent results from all three techniques is a much stronger argument for prioritizing FGFR2 than one technique alone.

What this analysis still cannot tell me

Predicted structures provide only a static snapshot without any information about metals, ligands, or other interacting molecules; that is why the hinge region of FGFR2 might act differently after interaction with real ligands. Additionally, structure predictions cannot tell how a cancer mutation will change a pocket's shape as it was in the case of p53 with Y220C. Resolving either of these issues will require an actual experimental study or a mutation resistance study of these two genes.

Third, selectivity is a limitation of my research which deserves mentioning in relation to FGFR2. It is well-known that FGFR2 belongs to a class of four highly homologous receptors, and their kinase domains, which contain the druggable pocket identified by me, are very similar in terms of structure. Any small molecule which was created as a ligand of the mentioned druggable pocket, could equally bind not only to FGFR2, but also to FGFR1, FGFR3, and FGFR4. Such an effect of binding to off-targets is one of the major causes of adverse reactions in medicines. Unfortunately, a predicted structure provides no information about selectivity of a future drug.

Checking my own reading against a known case

In order to validate this test method, I was able to confirm that the program identified p53 protein's crystal structure correctly at 0.68 Å.

Four residues, C176, H179, C238 and C242, are zinc ligands, while R248 and R273 directly contact DNA. R175, the hotspot I scored in Module 3, is neither, but it sits right next to the zinc site (5 Å), so it holds the structure together. That fits its Module 3 score being unusual but not extreme.

The broader lesson from p53 is one I tried to carry into my own targets rather than just note and move past. p53 is one of the most famous cancer genes there is, yet its own DNA-binding surface is broad, flat, and historically very hard to drug directly, which is a large part of why it has resisted targeted therapy for decades despite its importance. Reading that case correctly meant not assuming importance automatically equals druggability.

Applying that same caution to FGFR2 and CYP1A1 is why I leaned on the confidence rules and the ligand-site check rather than trusting a good pocket score by itself, the same discipline that explains why p53 remains a hard target even though it is arguably the most studied gene in all of cancer biology.

REF

1. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Žídek, A., Potapenko, A., Bridgland, A., Meyer, C., Kohl, S. A. A., Ballard, A. J., Cowie, A., Romera-Paredes, B., Nikolov, S., Jain, R., Adler, J., ... Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

2. Pacini, L., Jenks, A. D., Lima, N. C., & Huang, P. H. (2021). Targeting the fibroblast growth factor receptor (FGFR) family in lung cancer. Cells, 10(5), 1154. https://doi.org/10.3390/cells10051154